UNIVERSAL AI FILE MANAGER: AN MCP-BASED PLATFORM FOR INTELLIGENT MULTI-CLOUD AND LOCAL-FIRST HYBRID SEARCH AND RETRIEVAL

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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DECLARATION

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ABSTRACT

Modern digital workflows require users to manage files across disparate cloud providers (Google Drive, Microsoft OneDrive, Dropbox) and physical local drives (C:, D:, E:). Existing search utilities suffer from vendor fragmentation, inability to understand natural-language intent, and complete failure on multimodal visual assets where filenames are uninformative. Furthermore, transmitting private documents to remote centralized cloud indexing services exposes sensitive enterprise and personal data to severe security and compliance liabilities.

This project proposes and implements the **Universal AI File Manager**, an intelligent, privacy-preserving, local-first search and retrieval platform utilizing Anthropic's **Model Context Protocol (MCP)**. By decoupling file system tools into standardized MCP tool contracts (search_files, list_files, get_metadata, download_file), the platform establishes an interoperable interface that eliminates proprietary cloud API coupling. To ensure strict data sovereignty, a 100% on-device indexing and retrieval engine is implemented. The system combines lexical full-text indexing (SQLite FTS5), dense semantic text embeddings (all-MiniLM-L6-v2, 384 dimensions), and cross-modal visual embeddings (open_clip ViT-B/32, 512 dimensions) inside an embedded **LanceDB** vector database.

Candidates from the lexical, semantic, and visual retrieval pipelines are unified using **Reciprocal Rank Fusion (RRF, k=60)** with an NLP query parser that removes conversational carrier words, expands camelCase terms, and computes multi-term coverage. An Explainable AI (XAI) relevance matrix provides transparent attribution breakdown for every matched result with a user-facing toggle switch. Evaluated on a comprehensive benchmark of 20 diverse real-world fixtures across 30 natural-language queries, the platform achieves **100% Recall@1, 100% Recall@5, and a Mean Reciprocal Rank (MRR) of 1.000** with sub-50ms search latency.

Keywords: Model Context Protocol (MCP), Local-First AI, Hybrid Retrieval, Dense Vector Embeddings, Reciprocal Rank Fusion, Multimodal Search, OpenCLIP, SQLite FTS5, LanceDB, Explainable AI.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
BM25	Best Matching 25 (Probabilistic Lexical Ranking Algorithm)
CLIP	Contrastive Language-Image Pre-training
FTS5	Full-Text Search Version 5 (SQLite Engine)
GUI	Graphical User Interface
JWT	JSON Web Token
LLM	Large Language Model
MCP	Model Context Protocol (Anthropic Open Standard)
MRR	Mean Reciprocal Rank
NLP	Natural Language Processing
OCR	Optical Character Recognition
OAuth	Open Authorization Protocol (Version 2.0)
PKCE	Proof Key for Code Exchange
REST	Representational State Transfer
RRF	Reciprocal Rank Fusion
SHA-256	Secure Hash Algorithm 256-bit
SSE	Server-Sent Events
UI/UX	User Interface / User Experience
XAI	Explainable Artificial Intelligence

CHAPTER 1

INTRODUCTION

1.1 Overview of the Project

The proposed project focuses on designing and implementing the **Universal AI File Manager**, an intelligent, multi-cloud file aggregation and local-first semantic retrieval platform using the **Model Context Protocol (MCP)**. Modern computer users store thousands of documents, presentations, videos, and images across competing cloud storage services such as Google Drive, Microsoft OneDrive, and Dropbox, alongside local drive partitions (C:, D:, E:). Each storage provider operates within a separate commercial boundary, requiring individual authentication sessions, distinct query interfaces, and proprietary search operators.

The Universal AI File Manager replaces fragmented manual searching with a unified, conversational desktop platform. Rather than coupling application logic directly to vendor-specific APIs, the architecture adopts Anthropic's open-source **Model Context Protocol (MCP)**. An autonomous reasoning agent parses natural-language user queries into structured MCP tool requests. Concurrently, a 100% on-device indexing engine ingests documents and media files locally. Text embeddings are produced using all-MiniLM-L6-v2 (384-dimensional dense vectors), visual embeddings are produced using open_clip ViT-B/32 (512-dimensional vectors), and full-text keyword indices are maintained via SQLite FTS5. Candidate results are fused using Reciprocal Rank Fusion (RRF, k=60), guaranteeing rapid, private, and accurate retrieval.

1.2 Motivation of the Project

Two major paradigms motivate this research. First, enterprise and personal knowledge work has suffered significant productivity losses due to data silos. Users waste valuable time switching browser tabs to find files whose location is uncertain. Second, privacy concerns prohibit transmitting private documents—such as tax records, employment agreements, financial ledgers, and proprietary source code—to remote centralized cloud AI services for indexing. A practical file management architecture must deliver the semantic intelligence of modern vector search while guaranteeing absolute local data sovereignty where zero file contents or search queries ever leave the user's host workstation.

1.3 Problem Statement

Traditional desktop search utilities (such as Windows Search) and cloud provider search bars rely on brittle exact-string matching. They fail when users search conceptually (e.g., searching *'give me my latest resume'* when the file is named CV_Final_v3.pdf) or visually (e.g., searching *'autumn maple leaf'* when the photo is named IMG_4021.jpg). Furthermore, naive keyword matching frequently pollutes search results by matching conversational filler words (such as *'find'*, *'my'*, or *'year'*). This project addresses the challenge of building an interoperable, privacy-preserving, multi-cloud search system that resolves semantic, visual, and temporal queries with high precision.

1.4 Objectives of the Project

- To develop an interoperable multi-cloud file management architecture using Model Context Protocol (MCP SDK v2).
- To implement secure OAuth 2.0 PKCE authentication for Google Drive, Microsoft OneDrive, and Dropbox with Fernet AES-256 token encryption at rest.
- To build an on-device, offline hybrid AI search engine combining SQLite FTS5 (lexical) and LanceDB (dense vector).
- To implement zero-shot cross-modal visual search using open_clip ViT-B/32 without external cloud vision APIs.
- To formulate an NLP query parser eliminating conversational stopwords and handling camelCase tokenization.
- To develop an Explainable AI (XAI) relevance matrix and interactive user toggle switch justifying result rankings.

1.5 Scope of the Project

The project encompasses four storage connectors (Google Drive, OneDrive, Dropbox, and Windows local storage), supports all major file types (PDF, DOCX, TXT, MD, CSV, Code, PNG, JPG, MP4, MP3), and runs completely offline after initial local model initialization. Full integration is validated on desktop environments running Windows 10/11.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to the Domain

Information Retrieval (IR) has transitioned from Boolean keyword queries to dense vector semantic search powered by deep neural networks. While dense embeddings capture semantic similarity, lexical search remains superior for exact serial numbers, filenames, and acronyms. Hybrid search—combining inverted indexes with dense vector embeddings—has emerged as the gold standard in modern retrieval-augmented systems. Concurrently, Anthropic's Model Context Protocol (MCP) establishes an open industry standard for connecting foundation models to external data sources safely.

2.2 Literature Survey Comparison Table (Selected Studies)

No.	Paper / Authors, Year	Methodology	Advantages	Relevance to Project
1	Model Context Protocol Spec (Anthropic, 2024)	Standardized client-host-server tool calling protocol.	Decouples AI logic from proprietary APIs.	Forms core architecture of our system.
2	Dense Passage Retrieval (Karpukhin et al., 2020)	Dual-encoder dense representations for retrieval.	Captures conceptual meaning over keywords.	Informs dense semantic retrieval layer.
3	Learning Transferable Visual Models (Radford et al., 2021)	Joint multimodal contrastive pre-training (CLIP).	Enables zero-shot text-to-image retrieval.	Powers our visual image search engine.
4	Reciprocal Rank Fusion (Cormack et al., 2009)	Non-parametric rank score fusion formula.	Scale-invariant; avoids score skew.	Mathematical foundation for hybrid search.
5	Sentence-BERT (Reimers & Gurevych, 2019)	Siamese Transformer producing cosine embeddings.	Extremely fast inference on CPU (all-MiniLM).	Selected text embedding backbone.
6	SQLite FTS5 Spec (Hipp et al., 2015)	Inverted full-text index with BM25 ranking.	Sub-millisecond token matching on disk.	Selected lexical retrieval engine.
7	LanceDB Vector DB (LanceDB Team, 2023)	Embedded columnar vector database via Arrow.	Zero-server background overhead.	Embedded vector database engine.
8	LIME: Explainable AI (Ribeiro et al., 2016)	Local interpretable model-agnostic explanations.	Human-interpretable feature importance.	Informs relevance attribution panel.

2.3 Research Gaps Identified

- **Vendor API Lock-in:** Existing tools write custom brittle adapters for each cloud provider. *Response:* Standardized tool calling via MCP SDK v2.
- **Cloud Privacy Vulnerabilities:** Hosted AI search exposes confidential files. *Response:* Enforced 100% on-device embedding and indexing.
- **Multimodal Search Blindness:** Images without filenames are unsearchable. *Response:* Integrated open_clip ViT-B/32 text-to-image projections.
- **Conversational Keyword Over-matching:** Words like 'find' or 'my' trigger 100% false positives. *Response:* Engineered multi-term coverage query parser.

CHAPTER 3

PROBLEM DEFINITION AND PROPOSED METHODOLOGY

3.1 System Architecture

The Universal AI File Manager is architected across three decoupled tiers:

1. **Client Tier (React 18 / Vite):** Interactive single-page application providing a unified search bar, real-time cloud and local storage gauges, full file previewers, and an interactive 'Explain Relevance' switch.
2. **Application Core (FastAPI / Port 8000):** Orchestrates JWT auth, Fernet AES-256 token security, Ollama/Heuristic intent resolution, and the multi-factor ranking pipeline.
3. **Protocol & Search Tier (MCP Server / Port 8001):** Standardized MCP server exposing storage tools to the reasoning agent, coordinating SQLite FTS5 and LanceDB vector stores.

3.2 Mathematical and Retrieval Formulations

1. Dense Semantic Vector Cosine Similarity:

For query vector q and document chunk vector d (384 dimensions):

```
Sim_text(q, d) = (q · d) / (||q||_2 * ||d||_2) = (∑ q_j * d_j) / (sqrt(∑ q_j^2) * sqrt(∑ d_j^2))
```

2. Cross-Modal Text-to-Image Projections:

For text query T projected into CLIP joint space and visual image I (512 dimensions):

```
Sim_image(T, I) = (E_T(T) · E_I(I)) / (||E_T(T)||_2 * ||E_I(I)||_2)
```

3. Reciprocal Rank Fusion (RRF, k=60):

```
RRF Score(d) = ∑_{m ∈ methods} 1 / (60 + rank_m(d))
```

4. Multi-Factor Composite Scoring Formula:

```
Score_final(d) = 0.35 * S_kw + 0.35 * S_sem + 0.20 * S_rec + 0.10 * S_exact
```

Where temporal recency decay over age t in days is given by: S_rec(d) = 1.0 / (1.0 + (t / 90.0))

CHAPTER 4

REQUIREMENT ANALYSIS AND SYSTEM DESIGN

4.1 Hardware and Software Specifications

Component	Specification / Technology	Purpose
Processor	Intel Core i5 / AMD Ryzen 5 or higher	Data preprocessing and on-device model inference.
Memory (RAM)	8 GB minimum; 16 GB recommended	Concurrent loading of MiniLM and CLIP weights.
Disk Storage	5 GB SSD available space	LanceDB tables, SQLite DB, model checkpoints.
Backend Framework	Python 3.11 / 3.12, FastAPI, Uvicorn	High-performance asynchronous REST microservice.
MCP Protocol	Official Anthropic MCP Python SDK v2	Streamable HTTP tool dispatching.
Vector Database	LanceDB (Apache Arrow columnar)	Embedded, serverless vector persistence.
Frontend Framework	React 18, Vite, Tailwind CSS	Responsive dark-mode UI with live explainability.

4.2 Document Parsing and Ingestion Pipeline

Files from local directories and cloud providers are processed through a structured pipeline:

- **Text Extraction:** PyMuPDF extracts text streams from PDFs with RapidOCR fallback for scanned pages. DOCX files are parsed via python-docx; plain text, Markdown, and CSV are read directly.
- **Sliding-Window Chunking:** Long documents are segmented into chunks of ~400 tokens with 50-token overlap, prepended with logical filename and path headers.
- **Visual Embedding:** Image files are normalized and encoded via CLIP ViT-B/32 into 512-dimensional feature vectors.
- **Guaranteed Temporary Cleanup:** Cloud files downloaded temporarily for content indexing are deleted immediately inside a try...finally block, ensuring zero residual storage.

CHAPTER 5

PROJECT IMPLEMENTATION AND EVALUATION

5.1 Empirical Evaluation on 30 Natural-Language Benchmark Queries

To rigorously evaluate search precision, a test suite of 20 realistic benchmark fixtures (invoices, agreements, resumes, spreadsheets, code files, and photographs) was indexed into the hybrid retrieval engine. A suite of 30 complex natural-language queries was executed across the system:

Metric	Measured Score	Target Threshold	Evaluation Status
Total Queries Evaluated	30 Queries	30 Queries	100% Complete
Recall@1 (Top-1 Accuracy)	30 / 30 (100.0%)	>= 85.0%	PASSED
Recall@5 (Top-5 Accuracy)	30 / 30 (100.0%)	>= 95.0%	PASSED
Mean Reciprocal Rank (MRR)	1.000	>= 0.900	OPTIMAL
Average Search Latency	38.4 ms	< 100.0 ms	OPTIMAL

5.2 Selected Query Results Breakdown

Query Text	Target File	Top Match Result	Score
give me my latest resume	resume_latest_2026_ai_lead.pdf	resume_latest_2026_ai_lead.pdf	0.912
leaf png	green_oak_leaf.png	green_oak_leaf.png	0.894
autumn maple leaf	autumn_maple_leaf.jpg	autumn_maple_leaf.jpg	0.905
cloud hosting invoice	invoice_2026_01_cloud_hosting.pdf	invoice_2026_01_cloud_hosting.pdf	0.887
quarterly budget allocation	quarterly_budget_2025.csv	quarterly_budget_2025.csv	0.879
red sports car	sports_car_red.png	sports_car_red.png	0.918
finalDraft	finaldraft.mp4	finaldraft.mp4	0.825
DreamFolks	DreamFolks_Vid01.mp4	DreamFolks_Vid01.mp4	0.892
find my final year fees	Final_Year_Fees.txt	Final_Year_Fees.txt	0.895

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Work Completed

This project successfully developed and validated the Universal AI File Manager platform. Key engineering achievements include:

- Standardized tool execution layer connecting Google Drive, OneDrive, Dropbox, and Local drives using Model Context Protocol (MCP SDK v2).
- 100% on-device hybrid retrieval engine combining SQLite FTS5 (BM25), all-MiniLM-L6-v2 (dense text vectors), and open_clip ViT-B/32 (multimodal vision-language vectors).
- Reciprocal Rank Fusion (k=60) with multi-term NLP query cleaning, camelCase expansion, and compact token matching.
- Explainable AI (XAI) relevance matrix decomposing scores into keyword coverage, semantic similarity, and recency boosts with an interactive UI switch.
- Achieved 100% Recall@1, 100% Recall@5, and 1.000 MRR on benchmark evaluation queries.

6.2 Limitations and Future Enhancements

- **Local Keyframe Video Retrieval:** Future work will incorporate ffmpeg keyframe extraction to embed video scenes into LanceDB via CLIP.
- **On-Device Audio Transcription:** Integration of Whisper ONNX models will allow local indexing of voice recordings and podcasts.
- **Cross-Cloud Duplicate Elimination:** Cross-provider deduplication using SHA-256 hashes will enable single-click cloud storage cleanup.

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